Check for updates

Wiley
Journal of Advanced Transportation
Volume 2025, Article ID 9606650, 16 pages
https://doi.org/10.1155/atr/9606650

Research Article

Addressing Perceived Insecurity in Public Transportation: A PLS-SEM Approach

Carolina Busco,[1] **Felipe González**,[1] **Sara Arancibia**,[1] **Tiare Vera**,[1] **Milko Yuretic**,[1] **and Claudio Fuentes**[2]

[1]*Faculty of Engineering and Sciences, Diego Portales University, Santiago, Chile*
[2]*Faculty of Social Sciences, Diego Portales University, Santiago, Chile*

Correspondence should be addressed to Felipe González; felipe.gonzalezr@udp.cl

Received 7 May 2024; Accepted 2 April 2025

Academic Editor: Mohammad Rezwan Habib

Copyright © 2025 Carolina Busco et al. Journal of Advanced Transportation published by John Wiley & Sons Ltd. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

The sense of insecurity and fear of crime associated with public transportation are disadvantages that can deter individuals from using public transportation. We propose a methodology based on partial least squares structural equation modeling (PLS-SEM) to examine perceptions of insecurity among public transportation users in the Gran Valparaíso area and how these perceptions have influenced the corporate image of the services. To do so, we conducted a survey to assess perceptions of insecurity in situations involving harassment, crime, threatening social environments, crowded and isolated environments, and corporate image. Additionally, we considered gender and previous experiences of victimization in public transportation as moderating factors to identify significant differences among these groups. The main factor influencing the corporate image of buses and the metro is insecurity based on harassment. In terms of gender, women have a heightened perception of insecurity in crowded settings, influencing their feelings of insecurity based on harassment. Meanwhile, men's apprehension mainly relates to crime and is due to isolated environments.

Keywords: corporate image; gender; PLS-SEM; public transportation; security

1. Introduction

Over the years, policymakers have worried about ensuring public transportation safety and security. Safety and security are related but distinct concepts in the field of protection and risk management. Safety refers to the condition of being free from unintentional physical harm or accidents while using transportation services. This encompasses the absence of immediate physical risks due to internal factors such as human errors, technical failures, or unsafe conditions that could cause injury or bodily harm. On the other hand, security refers to protection against intentional threats and malicious actions that can cause physical, emotional, or psychological harm [1, 2].

Perceived security is defined as the extent to which a system aligns with users' security expectations, as reflected in their evaluations of the likelihood of encountering violent incidents such as theft, assaults, or harassment in public spaces [3, 4]. Some argue that perceived security holds greater significance than actual security [5], given its impact on passenger well-being and the social and economic development of the community [6].

The sense of insecurity and fear of crime associated with public transportation are disadvantages that can deter individuals from using public transportation [7], thereby influencing the volume of users and their travel patterns [8, 9]. Existing literature has demonstrated a strong connection between the sense of insecurity and the perceived service image [10]. However, the interrelationships between the factors that affect the perception of insecurity in public transportation vary depending on the context of the study, particularly in developing countries [11]. Consequently, it is

crucial to identify the factors contributing to users' perceptions of insecurity and how they influence the corporate image (IMA) of the public transportation system.

In this research, we propose a methodology based on partial least squares structural equation modeling (PLS-SEM) to measure the effect of different constructs (isolated environments [IE], overcrowded environments [OE], threatening social environments [TSE], harassment) on the general perception of insecurity (GPI) and how the latter influences the IMA of public transportation.

We empirically validate the proposed model using information obtained from a survey conducted among public transportation users in Gran Valparaíso, Chile (a conurbation made up of the commune of Valparaíso, and the surrounding communes: Viña del Mar, Quilpué, Villa Alemana, and Concón). Although several studies conducted in the country have shown that insecurity is one of the aspects that most influence the decision to use public transportation, the results apply to the capital, Santiago [12, 13]. It is important to study other regions of the country, for which there is little evidence on the subject.

Furthermore, our research integrates multiple constructs simultaneously to assess their combined impact on the overall perception of insecurity and the IMA of public transportation services. This comprehensive approach provides a more nuanced understanding of how various factors contribute to passengers' feelings of insecurity and how these perceptions influence their trust in public transportation providers.

By highlighting the importance of addressing security concerns to improve users' travel experience, our study underscores the potential to enhance the IMA through targeted strategies that mitigate passenger fears. Additionally, the inclusion of moderators such as gender and prior experiences of victimization offers a deeper insight into how perceptions of insecurity are constructed, particularly in urban contexts in Latin America, where this type of analysis has been limited.

Therefore, this research is intended to fill the existing gap regarding the effect of insecurity in public transportation in the specialized Latin American literature. The findings should support the development of comprehensive strategies to retain and attract new public transportation users in other urban areas.

The rest of this study is structured into six sections. In Section 2, we explore literature related to the constructs mentioned and highlight the primary hypotheses proposed for each one. Section 3 outlines the study's methodology and the proposed model. The outcomes of our model are presented in Section 4, while Section 5 is dedicated to discussing our research findings. Finally, in Section 6, we provide our conclusions.

2. Literature Review and Hypothesis

Researchers have used various techniques to study the perception of security in public transportation, such as probabilistic models [14, 15], factor analysis [16], or iterative semistructured survey methods such as the Delphi process [17].

However, an appropriate approach to working with concepts that measure perceptions, attitudes, and intentions is SEM due to their ability to model relationships between latent variables, which cannot be measured directly [18, 19]. These models have been used in several studies to measure public transportation passenger satisfaction, service quality, and loyalty, as well as IMA, safety, and security [20–22].

In studies that address the perception of insecurity in public transportation, for example, concepts such as "insecurity" or "harassment" are subjective variables that depend on users' perceptions. These perceptions are often influenced by multiple factors, such as the environment, social interaction, or previous experiences. The use of SEM allows simultaneous assessment of the direct and indirect relationships between these latent factors, providing an analysis of how various constructs affect perceptions or behaviors.

In this section, we conduct a bibliographic review that allows us to propose the hypotheses supporting the construction of the PLS-SEM.

2.1. IE. Studies have shown that passengers experience the highest level of apprehension at bus stops and stations, where staff presence is absent [23]. This unease can potentially escalate into criminal incidents and intensify the sense of fear among individuals [24].

The nature of the crimes experienced can be influenced by the number of individuals present within means of transport. For instance, theft or physical manipulation is more prone to happen in overcrowded conditions. Conversely, robbery, assault, or sexual assaults are more likely to occur in settings lacking social supervision [25]. Research from a gender perspective reveals that attacks against female travelers tend to occur in crowded environments [26, 27]. Nevertheless, investigations into women's security often concentrate on their apprehensions when traveling alone at night or in isolated areas [28, 29].

Valentine [30] identifies two specific types of spaces that can be particularly unsettling for women: enclosed spaces with limited exit options such as subway stations, and isolated spaces such as bus stops. The second scenario emphasizes that an empty transportation environment can generate unease, leading to the belief that no assistance may be available in an emergency [31].

Given this literature, it is pertinent to explore how the absence of passengers and public transportation personnel impacts both waiting environments and the use of public transportation, potentially affecting perceptions of insecurity based on crimes (IBC) and insecurity based on harassment (IBH).

2.2. OE. The invasion of one's personal space while using public transportation has been associated with heightened general anxiety [32]. Public transit often requires that strangers maintain close social proximity, usually reserved for personal relationships [33].

During rush hour, buses and trains with high passenger densities can create opportunities for inappropriate sexual behavior, such as groping and touching [34, 35]. Harassment

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

is primarily driven by overcrowding, as highlighted by Newton et al. [36]. This crowded environment can lead to discomfort for passengers who may experience unwanted physical contact, which would be considered inappropriate in most settings. Unfortunately, this also allows perpetrators to disguise their actions as socially acceptable [26].

In the literature, OEs are associated with increased perceptions of insecurity related to both crimes and harassment.

2.3. TSE. Studies suggest that specific social factors, such as noisy crowds, gangs, and individuals under the influence of alcohol, have the potential to increase an individual's perception of risk [37]. Female travelers might feel more anxious when it comes to issues such as public drunkenness, offensive language, verbal threats, and unwanted physical contact while using public transportation or waiting at bus stops [24].

Our research is part of a broader project encompassing a previous qualitative study that included focus groups and individual interviews with public transportation users. These qualitative phases revealed that such behaviors can result in discomfort and frustration throughout their journey and contribute to an increased sense of insecurity stemming from apprehensions about potential criminal activity or harassment.

This qualitative phase also raised the issue of informal activities around public transportation. In Chile and many other Latin American countries, it is common for people to engage in informal activities to make a living. These informal activities frequently entail boarding public transportation and seeking financial contributions from passengers due to artistic performances such as singing, dancing, acting, or selling various items. Alternatively, some individuals may ask for voluntary donations. This type of social interaction is sometimes well received by passengers but often perceived as threatening, which is why it is included as a variable in this construct.

Based on the literature and our qualitative findings, TSEs can influence passengers' perceptions of insecurity related to crimes and harassment.

2.4. IBC. Controlling the public transportation environment is challenging because it is accessible to everyone who pays the fare and even those who choose to evade it. Many of the potential victims of a crime are unsupervised and easily accessible. In addition, potential targets are often in crowded conditions or environments where it is difficult to monitor their belongings. Unlike street situations, where offenders may be unable to anticipate people's actions, in public transportation, behavioral options are more limited and, therefore, more predictable [38, 39].

The fear of criminal activity can considerably impact public transportation, potentially discouraging commuters from opting for urban transit due to its influence on their perception of personal security [15, 40]. Multiple studies suggest that the fear of crime wields a more substantial influence over public transportation utilization than the actual incidence of criminal events [41]. Additionally, research indicates that women tend to be more susceptible to heightened apprehensions regarding crime and are more inclined to adapt their behaviors in response to these concerns [24, 42, 43].

Furthermore, it is important to note that the waiting areas frequented by public transportation users, such as bus stops and train stations, tend to be more susceptible to criminal activities than other places [39, 44]. As a result, policies associated with public transportation planning should prioritize the establishment of more secure and pleasant surroundings for passengers throughout their travel experience [45, 46].

The literature suggests that IBC significantly affects the GPI among passengers.

2.5. IBH. Sexual harassment is defined as inappropriate conduct of a sexual nature [47], which can be categorized into three primary types: verbal, nonverbal, and physical [48]. The behaviors involved can vary from relatively mild to highly serious, encompassing a range that includes lewd comments, catcalls, ogling/leering, innuendos, sexual invitations, threats, displaying pornography, staring, being followed or photographed, masturbation, frotteurism, unwanted sexual touching, and rape [35].

Some authors differentiate sexual harassment into two categories: nonconfrontational and confrontational. The former includes behaviors such as suggestive looks and insults, while the latter pertains to physical actions such as following the victim, touching, and sexual assault [42, 49]. Within the realm of public transportation, sexual harassment primarily takes on a nonconfrontational nature, but the most frequent behaviors depend on the cultural elements of each country [50, 51].

Sexual harassment primarily affects women, and they frequently experience it in public spaces, including public transportation, as a prevalent form of violence against them [27, 50, 52]. Notably, women's apprehension of crime is often intertwined with their fear of sexual offenses [53, 54], and the high rates of sexual harassment encountered within transportation settings amplify the overall sense of insecurity [25].

Extensive literature supports that the fear of sexual harassment significantly impacts women's freedom of movement throughout their trip, considering variables such as the mode of transport, travel time, and selected routes, among other considerations [50, 51].

Therefore, IBH may impact both the perception of IBC and the GPI. Additionally, it may influence the IMA of the public transportation service.

2.6. GPI. Mahrous et al. [55] identified a strong correlation between a sense of security and the potential threats to one's security. Capasso da Silva and Rodrigues da Silva [3] proposed that this concept is rooted in an individual's perception of the likelihood of encountering violent incidents such as theft, assault, or harassment in public settings. Other apprehensions may include contracting diseases or traffic accidents [56].

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ni et al. [10] conducted a study on the connection between IMA and satisfaction in which security was considered an element of perceived quality, although not as a separate construct. Their research revealed a notable correlation between the IMA and the GPI. Similarly, Nguyen-Phuoc et al. [21] determined that the perceived safety and security of bus services in Vietnam directly influence their IMA. Although these studies adopted different approaches to explore the connection between these constructs, both concluded that there is a substantial and evident link between them.

The literature indicates that the GPI has a negative impact on the IMA of public transportation services.

2.7. IMA. Özkan et al. [57] discuss the concept of IMA as the mental picture consumers' hold of a company. Similarly, Sumaedi et al. [58] defined public transportation service as the mental photograph that passengers have of what is being offered, which includes symbolic meanings associated with a specific transportation service. This perceived image is a significant factor in marketing, as it can influence user satisfaction [10]. A positive IMA can attract new customers and instill trust in current ones, ultimately increasing customer satisfaction [59].

Existing research indicates that citizens are likelier to use public transportation services with a solid IMA [60]. The perception of image in public transportation services is mainly built based on the quality of the service [61]. Even so, we have sufficient evidence to hypothesize that perceptions of insecurity among transportation users in Gran Valparaíso affect the IMA. A service perceived as unsafe will reduce travelers' trust, worsening the perceived image. Conversely, if transportation is perceived as safe, trust in the public company will improve, enhancing its image. Public transportation service providers can attract new passengers and increase their income by improving their IMA [10].

Accordingly, we consider IMA a construct that reflects public transportation users' mental images through trust in and reputation of the service, and it is explained and influenced by all previously mentioned constructs.

Based on the literature reviewed above, we have formulated several hypotheses to be tested in our study. Table 1 summarizes these hypotheses, along with the relevant references, constructs involved, main findings from the cited studies, and the justification for each hypothesis.

3. Methodology

3.1. PLS-SEM Method. The current study employs SEM to explore the interconnected factors that influence passengers' perceptions of insecurity and image. Hair et al. [62] characterized SEM models as an advanced multivariate framework that links empirical data with theoretical constructs, integrating existing knowledge into an empirical analysis. In assessing SEM model parameters, two prominent methods are the covariance-based approach (CB-SEM) and the variance-based approach (PLS-SEM).

CB-SEM is intended to estimate parameters that best replicate the variance–covariance matrix by leveraging maximum likelihood and assumptions about data distribution. This approach is particularly suitable when the primary objective is the validation and confirmation of existing theories, as it allows for rigorous testing of hypothesized relationships within the model [62]. However, CB-SEM demands a substantial sample size with numerous variables and assumes that the data meet conditions of normality and homogeneity.

In contrast, PLS-SEM is intended to optimize the model's predictive capability for causal relationships while minimizing its residual variance. This methodology is better suited for exploratory research and does not require samples to adhere to a normal distribution [19, 62]. Furthermore, PLS-SEM requires fewer observations than CB-SEM to ensure statistical power, typically at least 100 observations [18].

Therefore, while PLS-SEM is advantageous for exploratory studies with smaller sample sizes and nonnormal data, CB-SEM is preferable when the researcher aims to confirm theoretical constructs with a large dataset that satisfies the necessary statistical assumptions [19, 62]. The choice between CB-SEM and PLS-SEM should be guided by the research objectives, sample characteristics, and data distribution, ensuring that the selected method aligns with the study's goals and the nature of the data collected.

Given the distinctions we have considered, the current study was based on the PLS-SEM approach, which is recommended when the research is exploratory or extends an established structural theory.

In line with the hypotheses outlined in Section 2, Figure 1 provides an overview of the conceptual model examined using the PLS-SEM methodology. In the diagram, the circles represent the constructs related to insecurity, while the arrows depict direct relationships between these constructs. Hypotheses featuring a positive sign signify positive direct relationships, whereas those with a negative sign indicate negative direct relationships.

In addition, we performed two multigroup analyses to explore whether there are differences in the elements that shape the perception of security, one according to the user's gender and one related to previous victimization experiences.

3.2. Survey and Study Area. To validate the proposed model, we surveyed commuters utilizing the bus and metro services in the Gran Valparaíso region. The urban bus network within the area comprises 2067 vehicles managed by nine distinct business units or companies. Bus fares are determined based on the traveled distance or section and the specific company, resulting in a highly intricate fare system. Conversely, the Valparaíso Metro links the communes of Valparaíso, Viña del Mar, Quilpué, and Villa Alemana, boasting a total of 20 stations spanning 43 km. The Valparaíso Metro employs an electronic card system for fares, considering user type, travel time (categorized as low,

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1: Summary of proposed hypotheses.

Hypothesis	Reference(s)	Construct	Main findings of the study	Justification of the hypothesis
H1a: The perception of isolated environments positively correlates with the perception of insecurity based on crimes.	Lynch and Atkins [23]; Yavuz and Welch [24]; Ding et al. [25]; Koskela and Pain [28]; Smith [29]; Valentine [30]; Loukaitou-Sideris [31]	Isolated environments (IE)	Absence of staff and passengers increases fear and vulnerability, leading to higher perception of crime risk.	Literature shows that isolated settings contribute to feelings of insecurity regarding crimes.
H1b: The perception of isolated environments positively correlates with the perception of insecurity based on harassment.	Same as H1a	Isolated environments (IE)	Same findings as H1a, but specifically affecting harassment perceptions.	Isolation increases vulnerability to harassment due to lack of witnesses or assistance.
H2a: The perception of overcrowded environments positively influences the perception of insecurity based on crimes.	Thomas [32]; Graham et al. [33]; Ceccato [34]; Gekoski et al. [35]; Newton et al. [36]; Ceccato and Paz [26]	Overcrowded environments (OE)	Overcrowding leads to discomfort, anxiety, and opportunities for crime disguised as normal interactions.	Overcrowded settings can increase the perceived risk of crimes such as theft or pickpocketing.
H2b: The perception of overcrowded environments positively influences the perception of insecurity based on harassment.	Same as H2a	Overcrowded environments (OE)	Overcrowding facilitates inappropriate sexual behaviors and harassment.	Literature indicates that harassment is often driven by overcrowding.
H3a: Threatening social environments positively influence the perception of insecurity based on crimes.	LaGrange et al. [37]; Yavuz and Welch [24]; Ceccato and Paz [26]	Threatening social environments (TSE)	Presence of gangs, intoxicated individuals, and informal activities increases perceived risk.	Threatening social behaviors heighten fear of crime among passengers.
H3b: Threatening social environments positively influence the perception of insecurity based on harassment.	Same as H3a	Threatening social environments (TSE)	Same findings as H3a, affecting harassment perceptions.	Threatening behaviors can be perceived as precursors to harassment incidents.
H4: Insecurity based on crime positively affects the general perception of insecurity.	Irvin-Erickson and La Vigne [38]; Smith and Clarke [39]; Hamid et al. [40]; Ingvardson and Nielsen [15]; Badiora et al. [41]; Gardner et al. [42]; Loukaitou-Sideris and Fink [43]; Yavuz and Welch [24]	Insecurity based on crime (IBC)	Fear of crime discourages public transportation use and increases overall insecurity perception.	Perceived crime risk directly impacts general feelings of insecurity.
H5a: Insecurity based on harassment positively impacts insecurity based on crime.	O'Donohue et al. [47]; Loukaitou-Sideris and Ceccato [48]; Gekoski et al. [35]; Clark et al. [27]; Kash [50]; Vera-Gray [52]; Ding et al. [25]	Insecurity based on harassment (IBH)	Harassment increases fear and affects perceptions of other crimes.	Harassment experiences can elevate concerns about crime in general.
H5b: Insecurity based on harassment positively impacts the general perception of insecurity.	Same as H5a	Insecurity based on harassment (IBH)	Harassment significantly impacts women's freedom of movement and overall insecurity.	Harassment contributes to higher overall insecurity perception.
H5c: Insecurity based on harassment negatively impacts the corporate image.	Lewis et al. [51]; Ni et al. [10]; Nguyen-Phuoc et al. [21]	Insecurity based on harassment (IBH)	Perceived insecurity affects trust and corporate image.	Negative experiences with harassment can deteriorate corporate image.
H6: General perception of insecurity has a negative impact on the corporate image.	Mahrous et al. [55]; Capasso da Silva and Rodrigues da Silva [3]; Ni et al. [10]; Nguyen-Phuoc et al. [21]	General perception of insecurity (GPI)	Strong correlation between security perceptions and corporate image.	Overall insecurity reduces trust and negatively affects corporate image.

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

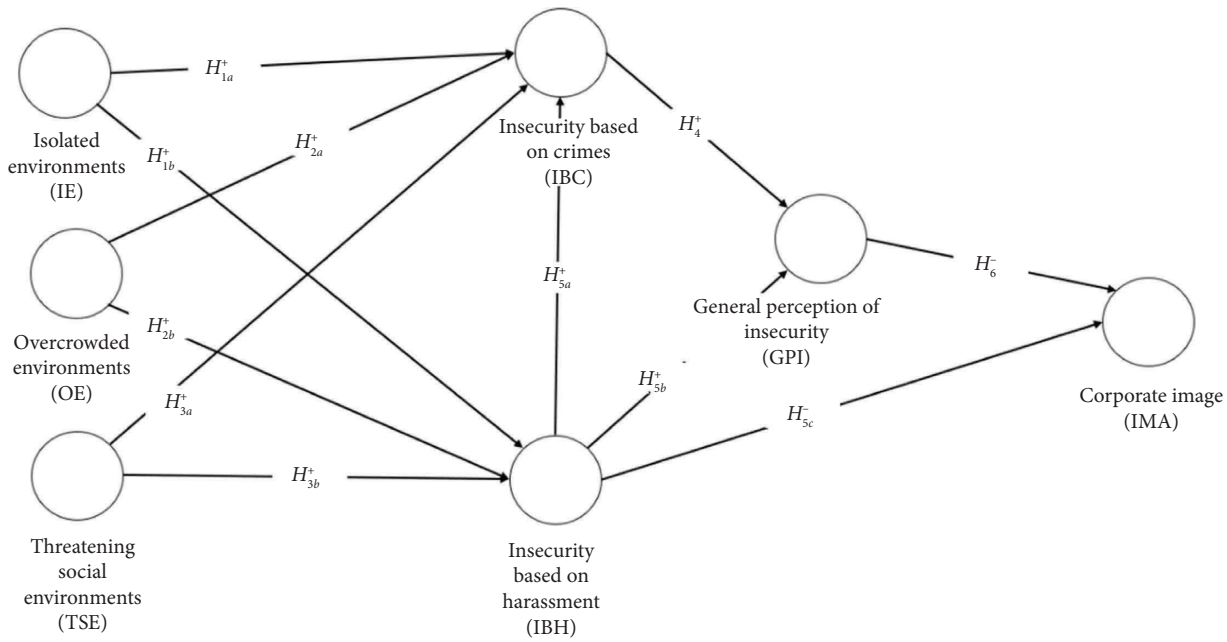

Figure 1: Proposed conceptual model.

medium, or high hour), and trip distance. This leads to a fare structure that varies across time slots, in which the only permanent variable is the distance covered. Currently, there is no fare integration for the various modes of transportation encompassed within the Gran Valparaíso area.

In terms of modal share, the 2014 Origin–Destination survey indicated that 38.61% of trips in the Gran Valparaíso area are made by public transportation. Urban buses account for 26.31% of these trips, while Metro Valparaíso accounts for around 2% and collective taxis contribute 7%. In contrast, private cars account for 29.43% of total trips, with 26.07% of trips being pedestrian. The region records approximately 2,295,100 trips per day [63].

However, the public transportation system in Gran Valparaíso faces several challenges that hinder its efficiency and accessibility. One of the main problems is the lack of fare integration between the different modes of transport, such as buses and the metro, which increases costs for users who depend on several modes to complete their trips. Furthermore, the system suffers from inadequate frequency and irregular service, especially in peripheral areas and during off-peak hours. High levels of congestion on key corridors further exacerbate the problem, resulting in longer travel times and overcrowded buses during peak periods. The region's mountainous topography also poses a challenge, limiting the expansion of bus routes and the implementation of specialized infrastructure such as dedicated bus lanes, which are critical to improving service regularity and reducing travel times [63].

The surveys were administered from May to June 2023 by seven trained pollsters. Surveying occurred in various locations, including waiting areas, buses, and urban trains within the Gran Valparaíso conurbation. The data were collected using traditional pencil and paper methods, and each survey took about 15 min to complete.

The surveys were conducted at different times of day to encourage participation from individuals of various age groups, and they were distributed across the five municipalities previously mentioned, as shown in Figure 2. The sample selection was nonprobabilistic and exploratory, following the criterion of equitable distribution according to gender, age, and commune based on data from the last census [64]. In total, 552 surveys were administered, and after the data purification process, 508 valid surveys were obtained.

The questionnaire was structured in three sections. The initial section was focused on gathering information about public transportation usage. The second section contained a set of questions designed on a seven-point Likert-type scale, ranging from "*completely disagree*" (1) to "*completely agree*" (7), to assess various statements from urban bus and train users. To ensure the clarity and comprehensibility of these statements, a pilot test of the questionnaire was conducted with a sample size of 30 individuals.

The final section of the questionnaire collected demographic information, including variables such as gender, age, nationality, sexual minority status, educational background, employment status, and access to personal transportation.

The design of the questionnaire was based on the literature review. Thus, eight items were created to measure IBC, IBH, and GPI. These items were adapted from previous studies that have explored these topics [12, 65, 66]. To assess perceptions of OE and IE, which have been shown to influence feelings of insecurity, four items were adapted from established scales in the work of Chowdhury and Van Wee [16] and Esmailpour et al. [65]. For IMA, four items were drawn from Nguyen-Phuoc et al. [21] to evaluate users' perceptions of the public transportation system in terms of trust and reputation. Finally, the construct of TSE, which

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

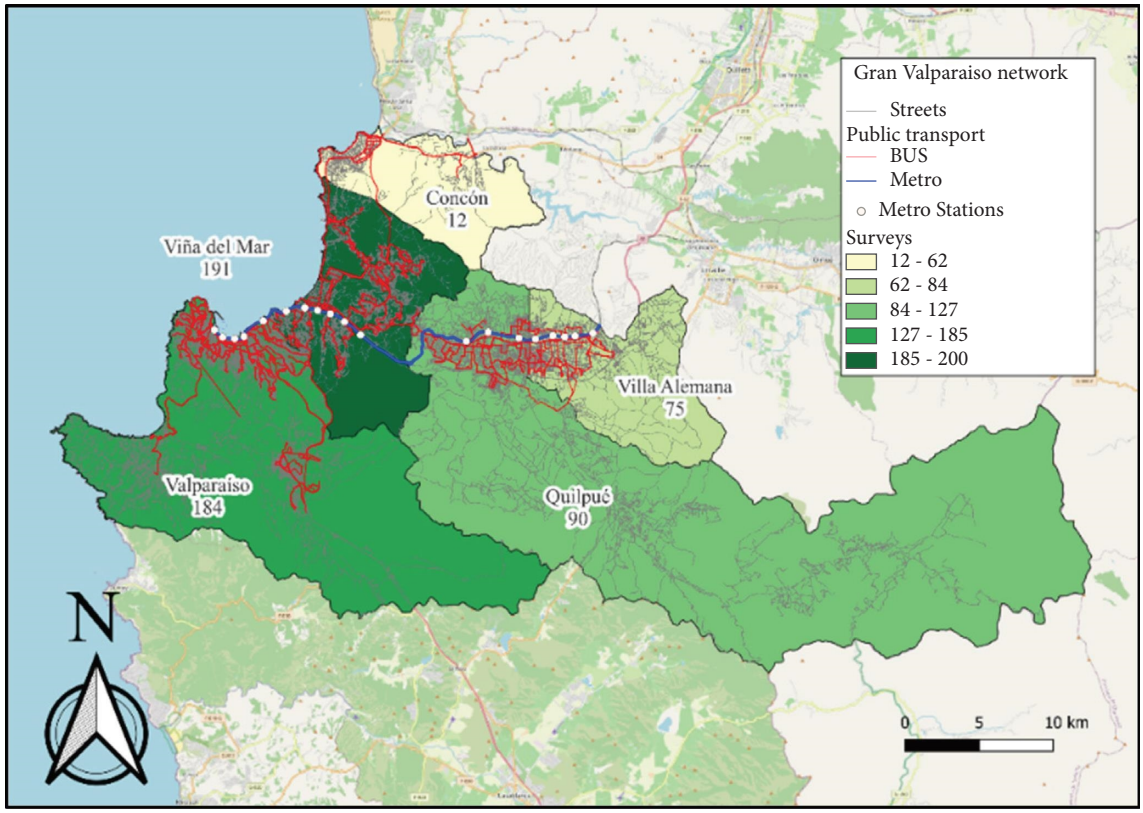

Figure 2: Distribution of surveys by commune of *Gran Valparaíso*.

includes variables such as the presence of gangs or public drunkenness, was measured using four items informed by user feedback from focus groups conducted during the qualitative phase of the research and existing literature, including the work of Kash [50]. Each construct's item with a Likert scale is presented in Table A1 (appendix).

4. Results

Due to the differences in operation between the bus and metro systems, two PLS-SEM models were built, one for each mode of transport. We present the findings in three main sections. First, we detail the outcomes derived from the questionnaire, providing insights into the profile and essential characteristics of the sample. Second, we outline the results of the PLS-SEM approach in three sequential steps as pursued in this study: fitting results of the measurement model, relevant results of the structural model, and total effect results. In the last section, we delve into the outcomes of the multigroup analysis.

4.1. Sample Profile. The primary sociodemographic characteristics of the sample can be outlined as follows. A total of 508 individuals participated in the study: 44.49% identified as male, 54.33% as female, and 0.79% as nonbinary. The largest age group represented was the 45–60 age range (25%), followed closely by the 25–34 age group (24.61%) and the 35–44 age group (20.28%). The smaller age groups included the 18–24 category (15.55%) and those over 60 (14.57%).

In terms of educational background, the majority (40.55%) had completed higher education, followed by those with incomplete higher education (27.76%) and individuals who had completed high school (24.01%). Regarding employment status, 42.32% worked full-time, 25.20% were students, and 12.60% worked part-time. Other categories included retirees (9.1%), stay-at-home moms/dads (5.31%), the unemployed (4.13%), and those not currently studying or working (1.38%). Additionally, the primary reason for using public transportation was for work purposes (47.44%), followed by commuting for educational reasons (25.79%) and personal errand travel (8.66%).

4.2. Results From the PLS-SEM Approach

4.2.1. Evaluation Measurement Models. The measurement model describes how the manifest variables, items, or indicators explain each latent variable. To evaluate the consistency of the constructs and their respective items, they must meet minimum conditions of validity and reliability. Table 2 presents the main indicators to carry out this validation.

To ensure precision, it is crucial to evaluate the degree to which each construct accurately accounts for the variance of its indicators, calculating the average variance extracted (AVE). Each construct has an AVE of at least 0.5, indicating that it explains 50% of the variance among its indicators, as recommended by Hair et al. [18]. To ensure that each concept is distinct, we evaluate the discriminant validity of

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2: Internal consistency reliability and convergent validity.

Construct	Indicator	Loading		CA		CR		AVE	
		Bus	Metro	Bus	Metro	Bus	Metro	Bus	Metro
Corporate image (IMA)	IMA1	0.870	0.883	0.882	0.897	0.908	0.925	0.711	0.756
	IMA2	0.750	0.812						
	IMA3	0.878	0.897						
	IMA4	0.869	0.882						
Isolated environments (IE)	IE1	0.884	0.930	0.706	0.823	0.872	0.919	0.773	0.850
	IE2	0.874	0.913						
Overcrowded environments (OE)	OE1	0.893	0.965	0.739	0.923	0.885	0.963	0.793	0.929
	OE2	0.888	0.962						
Threatening social environments (TSE)	TSE1	0.751	0.830	0.832	0.866	0.889	0.909	0.667	0.716
	TSE2	0.864	0.914						
	TSE3	0.854	0.890						
	TSE4	0.793	0.740						
Insecurity based on crime (IBC)	IBC1	0.888	0.887	0.779	0.795	0.873	0.881	0.698	0.711
	IBC2	0.878	0.868						
	IBC3	0.730	0.771						
Insecurity based on harassment (IBH)	IBH1	0.973	0.981	0.943	0.962	0.972	0.981	0.946	0.963
	IBH2	0.972	0.981						
General perception of insecurity (GPI)	GPI	1.000	1.000						

each one using the Heterotrait–Monotrait relationship, which measures the average correlation between indicators across constructs [67]. Finally, the model has discriminant validity because it meets values less than the required 0.85.

To assess the reliability of each variable, we determined how much variance is accounted for by the corresponding construct by calculating the square of the bivariate correlation between the indicator and the construct. According to Hair et al. [62]; this result should exceed 0.708, explaining over 50% of the indicator's variance. In Table 2, we can observe that all load values surpass the recommended threshold, confirming the reliability of the indicators.

Additionally, to ensure internal consistency and reliability within each construct, we compared the indicators associated with the same construct and assessed their association level. Several measurement indicators, such as Cronbach's alpha (CA) and composite reliability (CR), are available. Higher values of CA and CR indicate more robust reliability, with a recommended target value of over 0.7, as suggested by Nunnally [68]. Table 2 demonstrates that all constructs in our model exhibit values exceeding 0.7, signifying a reliable internal consistency among these constructs.

4.2.2. Evaluation of the Structural Model. Figures 3 and 4 show results for the PLS-SEM models for the bus and metro systems, respectively. In these models, relationships between several latent variables (represented by circles) are assessed. Each circle contains an R^2 value, which indicates the proportion of variance explained in that latent variable by its predictors. These latent variables include IE, OE, TSE, IBC, IBH, GPI, and IMA. Each latent variable is measured by observable indicators (rectangles) connected to them, which are used to assess the latent variable (the loading coefficient is reported in Table 2). The arrows between the constructs represent the hypothesized causal paths between the variables, and the numbers on these arrows indicate the coefficients of the paths—the strength and direction of the relationships. The values in parentheses next to the coefficients are p values, which indicate whether the relationships are statistically significant. Dashed arrows indicate nonsignificant relationships, while solid arrows represent significant relationships.

This procedure confirmed that nine hypotheses proposed in the theoretical framework were accepted for both the buses and metro cases (excluding H2a and H3a for buses and H2a and H6 for metro) at significance levels of 5%. On the other hand, in both models, the Q index was greater than 0 for all constructs except for the IMA in the metro model.

The results for the bus model reveal several significant relationships. For example, IE has a positive effect on IBC, with a path coefficient of 0.240 and a p value ≤ 0.001, indicating strong statistical significance. This indicates that passengers perceive a higher risk of crime when there is no supervision or presence of other passengers, which is consistent with previous studies associating loneliness at bus stops and inside buses with increased levels of vulnerability [23, 24]. Similarly, OE positively affect both IBC (0.171, $p = 0.007$) and IBH (0.206, $p = 0.001$). This suggests that high passenger density generates anxiety and increases opportunities for inappropriate behavior, a concern well-documented in public transportation literature [26, 36]. TSE strongly influence IBH (0.269, $p < 0.001$). In terms of mediation, GPI significantly mediates the effect of IBC on IMA, with a negative effect of GPI on IMA (−0.216, $p = 0.003$). Overall, the model suggests that higher perceptions of insecurity related to crime and harassment negatively affect the IMA of public transportation services, especially through the GPI, and that IBH shows a direct importance in shaping insecurity perceptions.

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

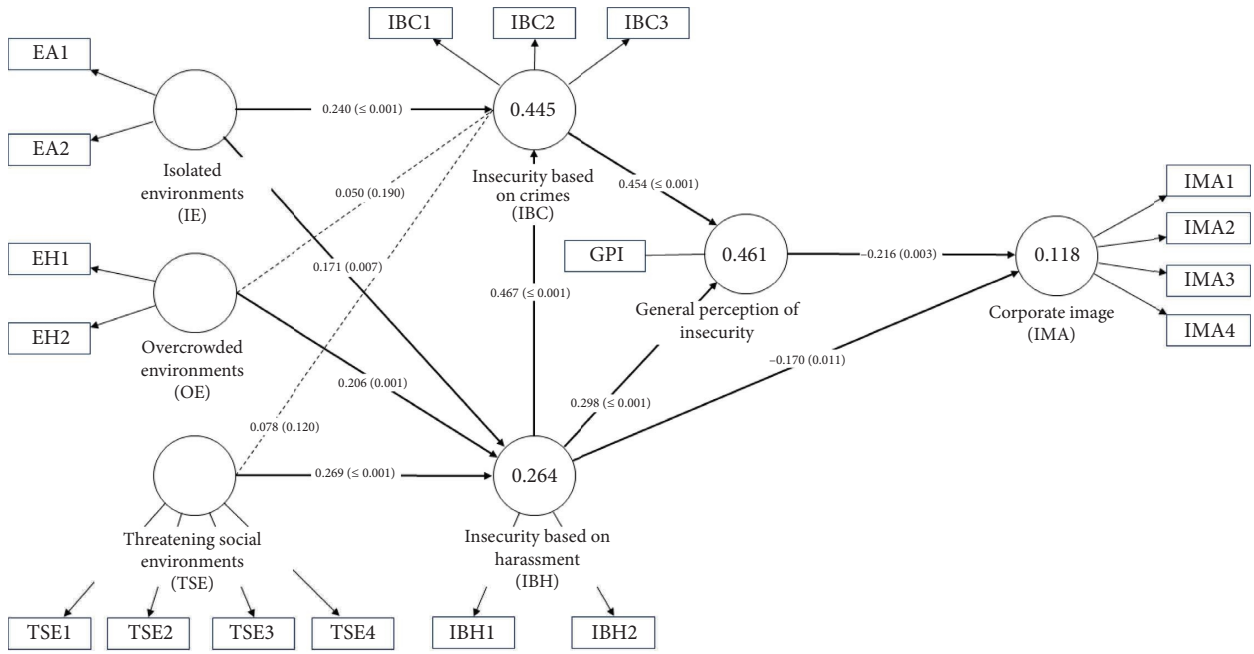

Figure 3: Results of the measurement and structural model for buses.

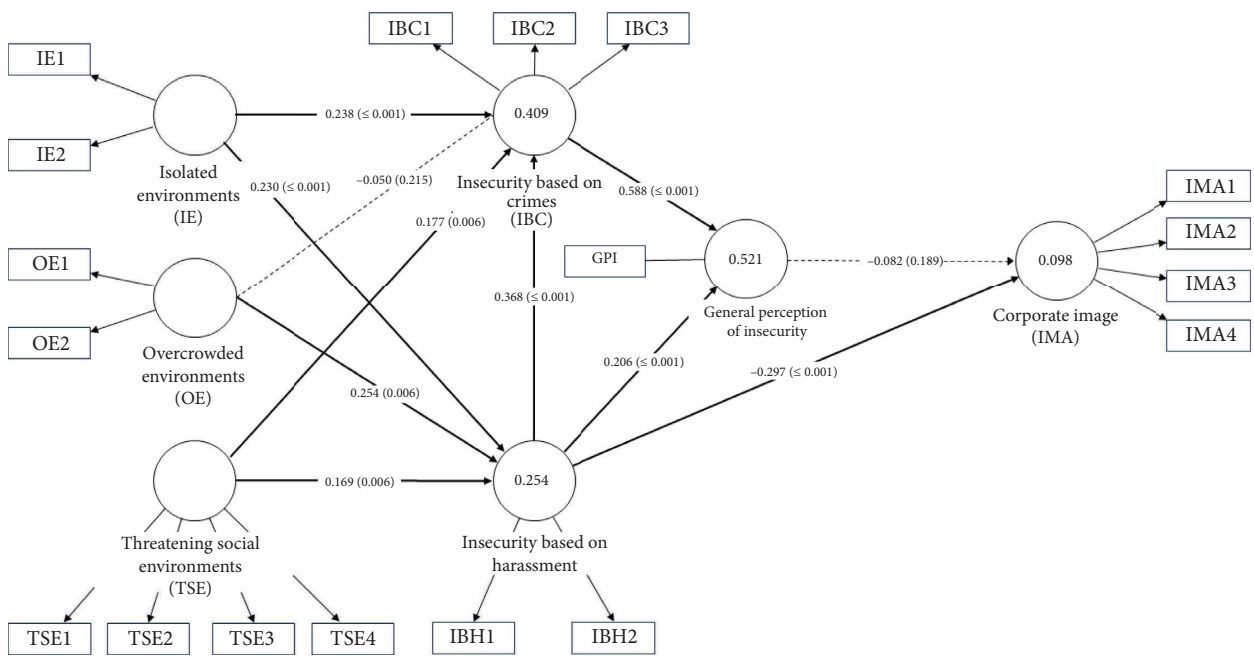

Figure 4: Measurement and structural model results for the metro.

The model for the metro shows several significant relationships between the constructs. For instance, IE has a positive and significant impact on IBC, with a path coefficient of 0.238 and a p value ≤ 0.001. However, unlike the bus system, IBC is more prominent in the metro, which may be related to the design of underground stations that creates a sense of isolation [30]. Similarly, OE significantly affect both IBC (0.177, $p = 0.006$) and IBH (0.254, $p = 0.006$). TSE also have a notable effect on IBH (0.169, $p = 0.006$). IBC strongly influences GPI, with a path coefficient of 0.368 and a p value ≤ 0.001, which in turn affects IMA, although this relationship is nonsignificant ($p = 0.189$). However, IBH negatively impacts IMA through GPI, with a path coefficient of −0.297 and a p value ≤ 0.001, reflecting that users prioritize personal security when evaluating the service [10, 21].

Finally, the obtained values for predictive relevance and R^2 surpass the accepted thresholds, effectively elucidating the extent of variance the model explains, except for IMA in the metro.

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

4.2.3. Total Effect. The total effect is the combined influence of direct and indirect effects as they propagate from one construct to another. Regarding the variables influencing the perception of insecurity, for buses, the perception of harassment has a total effect of 0.526. Meanwhile, perception of crimes is the most influential construct for the metro, with a value of 0.588, as shown in Table 3. This reveals that while crime is an important concern, harassment plays a fundamental role in shaping overall insecurity [27, 50].

In addition, IBH significantly affects the IMA of both buses and the metro. Furthermore, the negative impact of harassment on IMA (−0.283 for buses and −0.297 for the metro) highlights that harassment experiences erode trust in the transportation system, directly affecting the reputation of the services [10, 51]. The coefficients indicate that IBH is the most crucial factor to consider when creating a positive image of the service (refer to Table 4).

4.2.4. Multigroup Analysis. We conducted two multigroup analyses to investigate substantial distinctions based on two criteria: gender (male and female) and whether passengers had reported experiences of victimization (whether the user had been victimized on public transportation), using the bootstrap multigroup analysis technique with 5000 subsamples. This allowed us to identify variations in the beta coefficients for each group and report the p value of a two-tailed test. This analysis was intended to ensure that compositional invariance was maintained, at least to some extent, as outlined in Henseler et al. [69].

Table 5 reports the significant differences by gender for each mode of transportation. For buses, we discovered that the difference in coefficients between the relationship of IE ⟶ IBH and the disparity in the connection between IE ⟶ IBC shows a positive and highly significant impact at 5%. These effects manifest in coefficient differences of 0.276 and 0.260, respectively. In turn, the relationship of OE ⟶ IBH was shown to be negative and significant at 1%, with a coefficient of −0.352. On the other hand, regarding the metro, a significant difference is only found in the relationship of IBH ⟶ GPI, which is negative and has a coefficient of −0.276. This finding is consistent with the literature, which highlights that women often feel more vulnerable to sexual harassment, particularly in OE where incidents of unwanted touching or sexual assault may occur [50, 70].

Table 6 shows the significant differences related to people who reported experiences of victimization in public transportation compared to those who had not. In the case of buses, the relationships of IBC ⟶ GPI and GPI ⟶ IMA are significant and negative, with coefficients of −0.276 and −0.286, respectively. At the same time, the difference in the IE ⟶ IBC relationship is positive and significant at 10%. Finally, in the case of the metro, only the influence of OE ⟶ IBH is significant at 1%, obtaining a negative coefficient of −0.323, reflecting the findings of Sundling and Ceccato [71] that past experiences of victimization result in a lasting effect on future travel behaviors.

5. Discussion

To develop a sustainable and user-friendly public transportation system, it is necessary to facilitate the perception of security among its users. When a mode of transportation is perceived as insecure, it reduces the number of passengers; several authors have considered security a fundamental necessity for passengers [8, 72, 73]. This study has, therefore, expanded the body of knowledge by creating a theoretical model that assists in understanding the origins of passenger insecurity using a PLS-SEM approach with all constructs measured reflectively.

Our initial finding indicated that, for both modes of transportation (bus and metro), the perception of OE did not influence the IBC. However, it did show a significant association with IBH. This suggests that, among Gran Valparaíso users, high passenger density reduces concerns about becoming a victim of a crime, likely due to the increased social surveillance and a sense of companionship provided by fellow passengers. However, substantial OE, such as waiting areas and within public transportation, heighten concerns about harassment, such as unwanted touching or groping, aligning with existing literature on this topic [26, 34].

On the other hand, the relationship between TSE and IBC is rejected in the case of buses but not for the metro. This relationship was hypothesized due to the common occurrence of people engaging in informal activities within the public transportation network of Gran Valparaíso. This relationship proved to be significant for the metro, where it is much more common to find users engaging in these informal activities.

Another finding from our research is that, in the case of the GPI, the results vary depending on the mode of transportation. For buses, the most significant concern arises from the IBH, which was the most significant predictor of insecurity. The indicators show that this fear of being harassed is consistent both at bus stops and inside buses. Additionally, IBC was the second most influential predictor, with the primary indicator being the fear of experiencing a crime during the journey. Conversely, the opposite is true for the metro: the primary predictor of general insecurity is IBC, followed by IBH.

To better understand the relevance of IBH, it is worth highlighting some processes of social change in Chile that may shed light on this matter. Concerns about street harassment emerged within the framework of feminist demands from 2015 to 2017. Subsequently, in 2019, legislation was introduced to criminalize street harassment. In March 2023, the government initiated the "stand up" communication campaign targeting street harassment, particularly in the subway. Consequently, there is an awareness context wherein harassment has been a focal point of societal discourse in recent years, elucidating its prominence in the model.

In our research, we introduced the construct of the IMA to explore how it impacts passengers' confidence in a service. In the case of buses, we successfully validated a model that predicts image based on variables related to security, with

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3: Total effect on the *general perception of insecurity*.

Relation	Total effects		Standard deviation	
	Bus	Metro	Bus	Metro
IBH ⟶ GPI	0.526***	0.430***	0.052	0.059
IBC ⟶ GPI	0.454***	0.588***	0.075	0.059
TSE ⟶ GPI	0.143***	0.186***	0.041	0.050
IE ⟶ GPI	0.218***	0.242***	0.050	0.050
OE ⟶ GPI	0.110***	0.109***	0.036	0.033

***$p < 0.01$.

Table 4: Total effect on the *corporate image*.

Relation	Total effects		Standard deviation	
	Bus	Metro	Bus	Metro
IBH ⟶ IMA	−0.283***	−0.297***	0.054	0.049
IBC ⟶ IMA	−0.098***	−0.048	0.040	0.054
GPI ⟶ IMA	−0.216***	−0.082	0.078	0.093
TSE ⟶ IMA	−0.077***	−0.060***	0.026	0.025
IE ⟶ IMA	−0.076***	−0.080***	0.025	0.025
OE ⟶ IMA	−0.059***	−0.075***	0.022	0.024

***$p < 0.01$.

Table 5: Main gender differences.

Relation	Difference (male–female)	Two-tailed p value
Bus		
IE ⟶ IBH	0.276**	0.030
IE ⟶ IBC	0.260**	0.010
OE ⟶ IBH	−0.352***	0.003
Metro		
IBH ⟶ GPI	−0.276**	0.026

***$p < 0.01$.
**$p < 0.05$.

Table 6: Main differences in experiences of victimization.

Relation	Difference (no–yes)	Two-tailed p value
Bus		
IBC ⟶ GPI	−0.276**	0.029
GPI ⟶ IMA	−0.286**	0.036
IE ⟶ IBC	0.196*	0.058
Metro		
OE ⟶ IBH	−0.323***	0.005

***$p < 0.01$.
**$p < 0.05$.
*$p < 0.1$.

the most significant predictor being IBH. This result is consistent with specialized studies in the field, demonstrating that this insecurity influences users' mode of transportation choices and leads to a shift in passenger confidence [42, 50, 51]. In the case of the metro, we found that the GPI does not predict IMA, as the model's predictive and explanatory quality could not be validated.

Therefore, in our research, we found that the bus system is perceived as highly insecure by its passengers, leading to reduced trust in the service and significantly damaging its reputation. This finding aligns with the results of Nguyen-Phuoc et al. [21]. In contrast, the metro service is considered a relatively safe service. As a result, with a system that ensures passenger security, travelers prioritize other elements in shaping their perceived image of the metro service.

When including gender, we observed, on the one hand, one of the main factors contributing to men's apprehension about crime and harassment in buses is the presence of isolated bus stops or buses lacking surveillance, which fosters an environment conducive to potential criminal incidents.

On the other hand, for female metro users, their IBH has a noticeable impact on their perception of insecurity during their journeys. This outcome corresponds with existing literature, which has established that women experience a heightened fear of sexual harassment and sexual crimes [70]. Additionally, female users tend to feel more vulnerable in OE, which can potentially lead to incidents of unwanted touching or sexual assault against passengers. These findings are consistent with outcomes reported in gender-specific criminological studies [50].

We can additionally predict that, for those who have experienced episodes of victimization, feeling insecurity when traveling by bus is significantly more important in forming a negative image of the service. This is because such incidents lead to a loss of trust in the mode of transportation, thereby diminishing its perceived image.

Metro users who reported having experienced victimization in public transportation exhibited a heightened concern for IBH in OE, both in waiting areas and transportation modes, aligning with findings conducted by Sundling and Ceccato [71]. Their study revealed that individuals who have experienced victimization incidents in the past tend to feel less secure in the future. These findings are important to consider, as users who have been victims of sexual violence often alter their travel patterns after experiencing such incidents. They may opt for different modes of transportation, avoid traveling at specific

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

times of the day, and limit their exposure to public spaces. Additionally, these incidents tend to alter their state of mind in transportation settings, causing them to travel in a state of heightened vigilance [74].

6. Conclusions

In this research, we examined perceptions of insecurity among public transportation users in the Gran Valparaíso area and how these perceptions have influenced the IMA of the services. To do so, we conducted a survey to assess perceptions of insecurity in situations involving harassment, crime, TSE, IE and OE, and IMA. Additionally, gender and previous experiences of victimization in public transportation were also considered moderating factors to identify significant differences among these groups.

Considering that public transportation users should be seen as the primary recipients of public transportation policies [75], these findings allow for the formulation of recommendations to public transportation service providers. These recommendations are designed to mitigate perceived insecurity among passengers, thereby enhancing the overall image of their services. This, in turn, contributes to improving the efficacy of implementing decisions related to mobility management.

Findings revealed notable differences in the perception of insecurity between the bus and metro systems in the Gran Valparaíso area, with important implications for public transportation policy and practice. Results indicate that IBH is the primary factor shaping the perception of insecurity and the IMA of the bus system, while in the metro insecurity related to crimes has a greater influence. These results highlight the need for transport mode–specific security strategies: for buses, policies should prioritize combating sexual harassment, particularly at bus stops and inside buses. Practical measures could include the installation of security cameras, increased police presence, and awareness campaigns aimed at deterring harassment and educating passengers. For metro services, crime prevention strategies should focus on stations and intransit periods by improving surveillance and enhancing security infrastructure.

Additionally, our study emphasizes the significant gender disparities in perceptions of insecurity, with women reporting a heightened sense of vulnerability to harassment, especially in crowded environments such as bus stops and subways. These findings reinforce the necessity of gender-sensitive policies, including measures such as segregated waiting areas, designated safe zones in public transportation, and public awareness campaigns aimed at promoting respect for women in public spaces. Implementing these initiatives would help to reduce barriers that women face when using public transportation, making their journeys safer and more comfortable.

To mitigate perceptions of insecurity among public transportation users in the Gran Valparaíso region, it is essential to prioritize the components that contribute to the perception of harassment, as well as those contributing to the perception of crime in waiting. In this context, urban planners responsible for public transportation should place particular emphasis on the following aspects: (a) enhancing surveillance, involving transportation personnel, law enforcement, and the use of technological surveillance tools such as closed-circuit television (CCTV) at bus stops and platforms; (b) implementing crime prevention strategies through environmental design (CPTED), which includes measures such as improving visibility, lighting, alarms, accessible phones, and guaranteeing the proper maintenance of transportation facilities; and (c) providing alternative channels for reporting harassment such as text messaging and mobile applications that facilitate passenger tracking, documenting experiences, and creating maps of problematic areas [34]. Additionally, (d) referring to the relevance of the "stand up" campaign against street harassment and its possible effect on awareness, it is also advisable to recommend public campaigns of civic behavior in the transportation system (letting older people sit, harassment prevention, complaints channel, etc.) as well as promoting social surveillance.

Although this study has made valuable contributions, it is important to acknowledge its limitations. Although a multigroup analysis is conducted based on respondents' prior victimization experiences in public transportation, the precise nature of these experiences is not documented. It is fundamental to recognize that the impact of a theft differs from that of a violent robbery, or, for example, verbal harassment as distinct from sexual harassment. Additionally, we did not examine the time that had elapsed since the user's encounter with such events, and the fear generated may not have the same lasting effect over time.

Additionally, while our analysis was focused on gender, future studies could investigate the impact of other sociodemographic variables, such as age, education, or income, on perceptions of insecurity in public transportation. Additional analyses in these areas could provide deeper insights into how different population groups experience insecurity, leading to more inclusive policy interventions tailored to diverse needs.

In summary, our research underlines the importance of developing targeted public transportation policies that address the distinct perceptions of insecurity between bus and metro users, as well as the specific concerns of vulnerable groups, particularly women. By addressing harassment and crime with system-specific and gender-sensitive strategies, public transportation providers can not only enhance perceived safety but also increase ridership, fostering more inclusive, secure, and equitable urban mobility across Latin American cities.

Appendix A

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A1: Items by construct.

Construct	Indicator	Items
Corporate image (IMA)	IMA1	The bus/metro makes me feel confident.
	IMA2	The bus/metro has a better image than other transportation systems.
	IMA3	The bus/metro has a good reputation.
	IMA4	The overall image of the bus/metro is good.
Isolated environments (IE)	IE1	I feel insecure when there are no people at the metro platform or bus stop.
	IE2	I feel insecure when there are no people in the metro car or inside the bus.
Overcrowded environments (OE)	OE1	I feel insecure when there are too many people at the metro platform or bus stop.
	OE2	I feel insecure when there are too many people in the metro car or inside the bus.
Threatening social environments (TSE)	TSE1	I feel insecure when a person (or group) approaches me persistently to interact.
	TSE2	I feel insecure when a person (or group) persistently tries to sell me something.
	TSE3	I feel insecure when a person (or group) persistently asks for money.
	TSE4	I feel insecure when someone stands too close, even though there is plenty of space to move away.
Insecurity based on crime (IBC)	IBC1	I worry about being a victim of crime when I travel by bus/metro.
	IBC2	I worry about being a victim of crime when waiting for the bus/metro.
	IBC3	I worry when the bus goes through a neighborhood with a bad reputation/I worry when I don't see security guards in the metro.
Insecurity based on harassment (IBH)	IBH1	I am concerned about being harassed at the metro platform or bus stop.
	IBH2	I worry about being harassed on the metro car or inside the bus.
General perception of insecurity (GPI)	GPI	In general, I feel insecure when traveling on the bus/metro.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Funding

C.B. and F.G. acknowledge Proyecto Asociativo UDP 2023 and FONDECYT grant number 1230473. C.F. acknowledges FONDECYT 1240231 and Exploración Anid 13220187.

Acknowledgments

C.B. and F.G. acknowledge Proyecto Asociativo UDP 2023 and FONDECYT grant number 1230473. C.F. acknowledges FONDECYT 1240231 and Exploración Anid 13220187.

References

[1] V. Ceccato and A. Newton, *Safety and Security in Transit Environments: An Interdisciplinary Approach* (Springer, 2015).

[2] T. B. Joewono and H. Kubota, "Safety and Security Improvement in Public Transportation Based on Public Perception in Developing Countries," *IATSS Research* 30, no. 1 (2006): 86–100, https://doi.org/10.1016/s0386-1112(14)60159-x.

[3] D. Capasso da Silva and A. N. Rodrigues da Silva, "Sustainable Modes and Violence: Perceived Safety and Exposure to Crimes on Trips to and from a Brazilian University Campus," *Journal of Transport & Health* 16 (2020): 100817, https://doi.org/10.1016/j.jth.2019.100817.

[4] V. Ceccato, N. Gaudelet, and G. Graf, "Crime and Safety in Transit Environments: A Systematic Review of the English and the French Literature, 1970–2020," *Public Transport* 14, no. 1 (2022): 105–153, https://doi.org/10.1007/s12469-021-00265-1.

[5] H. Iseki and B. D. Taylor, "Style Versus Service? an Analysis of User Perceptions of Transit Stops and Stations," *Journal of Public Transportation* 13, no. 3 (2010): 23–48, https://doi.org/10.5038/2375-0901.13.3.2.

[6] F. Greco and A. Polli, "Security Perception and People Well-Being," *Social Indicators Research* 153, no. 2 (2020): 741–758, https://doi.org/10.1007/s11205-020-02341-8.

[7] A. Church, M. Frost, and K. Sullivan, "Transport and Social Exclusion in London," *Transport Policy* 7, no. 3 (2000): 195–205, https://doi.org/10.1016/s0967-070x(00)00024-x.

[8] P. Coppola and F. Silvestri, "Gender Inequality in Safety and Security Perceptions in Railway Stations," *Sustainability* 13, no. 7 (2021): 4007, https://doi.org/10.3390/su13074007.

[9] D. K. Kacharo, E. Teshome, and T. Woltamo, "Safety and Security of Women and Girls in Public Transport," *Urban, planning and transport research* 10, no. 1 (2022): 1–19, https://doi.org/10.1080/21650020.2022.2027268.

[10] A. Ni, C. Zhang, Y. Hu, W. Lu, and H. Li, "Influence Mechanism of the Corporate Image on Passenger Satisfaction With Public Transport in China," *Transport Policy* 94 (2020): 54–65, https://doi.org/10.1016/j.tranpol.2020.04.014.

[11] J. Mandhani, J. K. Nayak, and M. Parida, "Interrelationships Among Service Quality Factors of Metro Rail Transit System: An Integrated Bayesian Networks and PLS-SEM Approach," *Transportation Research Part A: Policy and Practice* 140 (2020): 320–336, https://doi.org/10.1016/j.tra.2020.08.014.

[12] C. Busco, F. González, P. Carmona, A. Cancino, and B. Alonso, "Social Exclusion and the Public Bus System in Santiago, Chile," *Cities* 132 (2023): 104092, https://doi.org/10.1016/j.cities.2022.104092.

[13] L. Sagaris and I. Tiznado-Aitken, *Sustainable Transport and Gender Equity: Insights From Santiago, Chile', Urban Mobility*

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and Social Equity in Latin America: Evidence, Concepts, Methods (Emerald Publishing Limited, 2020).

[14] R. F. Abenoza, V. Ceccato, Y. O. Susilo, and O. Cats, "Individual, Travel, and Bus Stop Characteristics Influencing Travelers' Safety Perceptions," *Transportation Research Record: Journal of the Transportation Research Board* 2672, no. 8 (2018): 19–28, https://doi.org/10.1177/0361198118758677.

[15] J. B. Ingvardson and O. A. Nielsen, "The Influence of Vicinity to Stations, Station Characteristics and Perceived Safety on Public Transport Mode Choice: A Case Study From Copenhagen," *Public Transport* 14, no. 2 (2021): 459–480, https://doi.org/10.1007/s12469-021-00285-x.

[16] S. Chowdhury and B. Van Wee, "Examining Women's Perception of Safety During Waiting Times at Public Transport Terminals," *Transport Policy* 94 (2020): 102–108, https://doi.org/10.1016/j.tranpol.2020.05.009.

[17] S. Cafiso, A. Di Graziano, and G. Pappalardo, "Using the Delphi Method to Evaluate Opinions of Public Transport Managers on Bus Safety," *Safety Science* 57 (2013): 254–263, https://doi.org/10.1016/j.ssci.2013.03.001.

[18] J. Hair, J. F. Hair Jr, G. T. M. Hult, C. M. Ringle, and M. Sarstedt, *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)* (Sage publications, 2021).

[19] J. Henseler, G. Hubona, and P. A. Ray, "Using PLS Path Modeling in New Technology Research: Updated Guidelines," *Industrial Management and Data Systems* 116, no. 1 (2016): 2–20, https://doi.org/10.1108/imds-09-2015-0382.

[20] P.-F. Chou, C.-S. Lu, and Y.-H. Chang, "Effects of Service Quality and Customer Satisfaction on Customer Loyalty in High-Speed Rail Services in Taiwan," *Transportmetrica: Transport Science* 10, no. 10 (2014): 917–945, https://doi.org/10.1080/23249935.2014.915247.

[21] D. Q. Nguyen-Phuoc, A. T. Phuong Tran, T. V. Nguyen, P. T. Le, and D. N. Su, "Investigating the Complexity of Perceived Service Quality and Perceived Safety and Security in Building Loyalty Among Bus Passengers in Vietnam–A PLS-SEM Approach," *Transport Policy* 101 (2021): 162–173, https://doi.org/10.1016/j.tranpol.2020.12.010.

[22] C. Zhang, Y. Liu, W. Lu, and G. Xiao, "Evaluating Passenger Satisfaction Index Based on PLS-SEM Model: Evidence From Chinese Public Transport Service," *Transportation Research Part A: Policy and Practice* 120 (2019): 149–164, https://doi.org/10.1016/j.tra.2018.12.013.

[23] G. Lynch and S. Atkins, "The Influence of Personal Security Fears on Women's Travel Patterns," *Transportation* 15, no. 3 (1988): 257–277, https://doi.org/10.1007/bf00837584.

[24] N. Yavuz and E. W. Welch, "Addressing Fear of Crime in Public Space: Gender Differences in Reaction to Safety Measures in Train Transit," *Urban Studies* 47, no. 12 (2010): 2491–2515, https://doi.org/10.1177/0042098009359033.

[25] H. Ding, A. Loukaitou-Sideris, and A. W. Agrawal, "Sexual Harassment and Assault in Transit Environments: A Review of the English-language Literature," *Journal of Planning Literature* 35, no. 3 (2020): 267–280, https://doi.org/10.1177/0885412220911129.

[26] V. Ceccato and Y. Paz, "Crime in São Paulo's Metro System: Sexual Crimes Against Women," *Crime Prevention and Community Safety* 19, no. 3-4 (2017): 211–226, https://doi.org/10.1057/s41300-017-0027-2.

[27] S. K. Clark, E. L. Jeglic, C. Calkins, and J. R. Tatar, "More Than a Nuisance: The Prevalence and Consequences of Frotteurism and Exhibitionism," *Sexual Abuse* 28, no. 1 (2016): 3–19, https://doi.org/10.1177/1079063214525643.

[28] H. Koskela and R. Pain, "Revisiting Fear and Place: Women's Fear of Attack and the Built Environment," *Geoforum* 31, no. 2 (2000): 269–280, https://doi.org/10.1016/s0016-7185(99)00033-0.

[29] M. J. Smith, "Addressing the Security Needs of Women Passengers on Public Transport," *Security Journal* 21, no. 1-2 (2008): 117–133, https://doi.org/10.1057/palgrave.sj.8350071.

[30] G. Valentine, "Women's Fear and the Design of Public Space," *Built Environment* no. 1978 (1990): 288–303.

[31] A. Loukaitou-Sideris, "Fear and Safety in Transit Environments From the Women's Perspective," *Security Journal* 27, no. 2 (2014): 242–256, https://doi.org/10.1057/sj.2014.9.

[32] J. A. P. K. Thomas, "'The Social Environment of Public Transport" (Doctoral dissertation, Open Access Te Herenga Waka-Victoria University of Wellington, 2009).

[33] C. Graham, A. Delbosc, and S. Mahmound, "Factors Influencing Young Peoples Perceptions of Personal Safety on Public Transport," *Journal of Public Transportation* 1, no. 1 (2013).

[34] V. Ceccato, "Women's Victimisation and Safety in Transit Environments," *Crime Prevention and Community Safety* 19, no. 3-4 (2017): 163–167, https://doi.org/10.1057/s41300-017-0024-5.

[35] A. Gekoski, J. M. Gray, M. A. Horvath, S. Edwards, A. Emirali, and J. R. Adler, "What Works' in Reducing Sexual Harassment and Sexual Offences on Public Transport Nationally and Internationally: A Rapid Evidence Assessment" (2015).

[36] A. D. Newton, H. Partridge, and A. Gill, "Above and below: Measuring Crime Risk in and Around Underground Mass Transit Systems," *Crime science* 3, no. 1 (2014): 1–14, https://doi.org/10.1186/2193-7680-3-1.

[37] R. L. LaGrange, K. F. Ferraro, and M. Supancic, "Perceived Risk and Fear of Crime: Role of Social and Physical Incivilities," *Journal of Research in Crime and Delinquency* 29, no. 3 (1992): 311–334, https://doi.org/10.1177/0022427892029003004.

[38] Y. Irvin-Erickson and N. La Vigne, "A Spatio-Temporal Analysis of Crime at Washington, DC Metro Rail: Stations' Crime-Generating and Crime-Attracting Characteristics as Transportation Nodes and Places," *Crime science* 4 (2015): 14–13, https://doi.org/10.1186/s40163-015-0026-5.

[39] M. J. Smith and R. V. Clarke, "Crime and Public Transport," *Crime and Justice* 27 (2000): 169–233, https://doi.org/10.1086/652200.

[40] N. A. Hamid, P.-L. Tan, M. F. M. Zali, N. U. Rahamat, and N. Abd Aziz, "Safety and Security Needs of Commuter Rail Services-Travellers' Perceptions," *Journal of the Eastern Asia Society for Transportation Studies* 11 (2015): 1495–1506.

[41] A. I. Badiora, O. S. Ojewale, and O. H. Okunola, "Perceived Risk and Fear of Crime in Public Transport Nodes: The Experience From Nigerian Transit Environment," *International Journal of Criminal Justice Sciences* 10, no. 2 (2015).

[42] N. Gardner, J. Cui, and E. Coiacetto, "Harassment on Public Transport and Its Impacts on Women's Travel Behaviour," *Australian Planner* 54, no. 1 (2017): 8–15, https://doi.org/10.1080/07293682.2017.1299189.

[43] A. Loukaitou-Sideris and C. Fink, "Addressing Women's Fear of Victimization in Transportation Settings: A Survey of US Transit Agencies," *Urban Affairs Review* 44, no. 4 (2009): 554–587, https://doi.org/10.1177/1078087408322874.

[44] R. Liggett, A. Loukaitou-Sideris, and H. Iseki, "Bus Stop–Environment Connection: Do Characteristics of the Built Environment Correlate With Bus Stop Crime?" *Transportation Research Record: Journal of the Transportation*

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Research Board 1760, no. 1 (2001): 20–27, https://doi.org/10.3141/1760-03.

[45] V. Ceccato, *Safety on the Move: Crime and Perceived Safety in Transit Environments* (Springer, 2014).

[46] P. Cozens and T. van der Linde, "Perceptions of Crime Prevention Through Environmental Design (CPTED) at Australian Railway Stations," *Journal of Public Transportation* 18, no. 4 (2015): 73–92, https://doi.org/10.5038/2375-0901.18.4.5.

[47] W. O'Donohue, K. Downs, and E. A. Yeater, "Sexual Harassment: A Review of the Literature," *Aggression and Violent Behavior* 3, no. 2 (1998): 111–128, https://doi.org/10.1016/s1359-1789(97)00011-6.

[48] A. Loukaitou-Sideris and V. Ceccato, "Sexual Harassment on Transit: A Global, Comparative Examination," *Security Journal* 35, no. 1 (2022): 175–204, https://doi.org/10.1057/s41284-020-00271-1.

[49] M. M. Davidson, M. S. Butchko, K. Robbins, L. W. Sherd, and S. J. Gervais, "The Mediating Role of Perceived Safety on Street Harassment and Anxiety," *Psychology of Violence* 6, no. 4 (2016): 553–561, https://doi.org/10.1037/a0039970.

[50] G. Kash, "Always on the Defensive: The Effects of Transit Sexual Assault on Travel Behavior and Experience in Colombia and Bolivia," *Journal of Transport & Health* 13 (2019): 234–246, https://doi.org/10.1016/j.jth.2019.04.004.

[51] S. Lewis, P. Saukko, and K. Lumsden, "Rhythms, Sociabilities and Transience of Sexual Harassment in Transport: Mobilities Perspectives of the London Underground," *Gender, Place & Culture* 28, no. 2 (2021): 277–298, https://doi.org/10.1080/0966369x.2020.1734540.

[52] F. Vera-Gray, "Men's Stranger Intrusions: Rethinking Street Harassment," *Women's Studies International Forum* (Elsevier, 2016).

[53] H.-P. Hsu, "How Does Fear of Sexual Harassment on Transit Affect Women's Use of Transit?" *Women's Issues in Transportation* 2 (2010): 85–94.

[54] D. C. May, N. E. Rader, and S. Goodrum, "A Gendered Assessment of the "threat of Victimization": Examining Gender Differences in Fear of Crime, Perceived Risk, Avoidance, and Defensive Behaviors," *Criminal Justice Review* 35, no. 2 (2010): 159–182, https://doi.org/10.1177/0734016809349166.

[55] A. M. Mahrous, Y. M. Moustafa, and M. A. Abou El-Ela, "Physical Characteristics and Perceived Security in Urban Parks: Investigation in the Egyptian Context," *Ain Shams Engineering Journal* 9, no. 4 (2018): 3055–3066, https://doi.org/10.1016/j.asej.2018.07.003.

[56] M. Friman, K. Lättman, and L. E. Olsson, "Public Transport Quality, Safety, and Perceived Accessibility," *Sustainability* 12, no. 9 (2020): 3563, https://doi.org/10.3390/su12093563.

[57] P. Özkan, S. Süer, İ.K. Keser, and İ.D. Kocakoç, "The Effect of Service Quality and Customer Satisfaction on Customer Loyalty: The Mediation of Perceived Value of Services, Corporate Image, and Corporate Reputation," *International Journal of Bank Marketing* 38, no. 2 (2020): 384–405.

[58] S. Sumaedi, I. G. M. Y. Bakti, T. Rakhmawati, N. J. Astrini, T. Widianti, and M. Yarmen, "Factors Influencing Public Transport Passengers' Satisfaction: A New Model," *Management of Environmental Quality: An International Journal* 27, no. 5 (2016): 585–597, https://doi.org/10.1108/meq-05-2015-0084.

[59] H.-H. Hu, J. Kandampully, and T. D. Juwaheer, "Relationships and Impacts of Service Quality, Perceived Value, Customer Satisfaction, and Image: An Empirical Study," *Service Industries Journal* 29, no. 2 (2009): 111–125, https://doi.org/10.1080/02642060802292932.

[60] C. W. Kuo and M. L. Tang, "Relationships Among Service Quality, Corporate Image, Customer Satisfaction, and Behavioral Intention for the Elderly in High Speed Rail Services," *Journal of Advanced Transportation* 47, no. 5 (2013): 512–525, https://doi.org/10.1002/atr.179.

[61] A. N. H. Ibrahim and M. N. Borhan, "The Interrelationship Between Perceived Quality, Perceived Value and User Satisfaction towards Behavioral Intention in Public Transportation: A Review of the Evidence," *International Journal of Advanced Science, Engineering and Information Technology* 10, no. 5 (2020): 2048–2056, https://doi.org/10.18517/ijaseit.10.5.12818.

[62] J. F. Hair, C. M. Ringle, and M. Sarstedt, "Partial Least Squares Structural Equation Modeling: Rigorous Applications, Better Results and Higher Acceptance," *Long Range Planning* 46, no. 1-2 (2013): 1–12, https://doi.org/10.1016/j.lrp.2013.01.001.

[63] Sectra, "Master Plan for Public Transport in Greater Valparaíso," (2021), https://www.sectra.gob.cl/planes_transporte_urbano/valparaiso/documentos_valparaiso.html.

[64] INE, "2017 Population and Housing Census: Preliminary Results," (2017), https://resultados.censo2017.cl/.

[65] J. Esmailpour, K. Aghabayk, M. Aghajanzadeh, and C. De Gruyter, "Has COVID-19 Changed Our Loyalty towards Public Transport? Understanding the Moderating Role of the Pandemic in the Relationship between Service Quality, Customer Satisfaction and Loyalty," *Transportation Research Part A: Policy and Practice* 162 (2022): 80–103, https://doi.org/10.1016/j.tra.2022.05.023.

[66] Y. Yang, S. Hu, D. Liao, and X. Huang, "What Affects Safety Perception of Female Ride-Hailing Passengers? An Empirical Study in China Context," *Journal of Advanced Transportation* 2022 (2022): 1–16, https://doi.org/10.1155/2022/3316535.

[67] J. Henseler, C. M. Ringle, and M. Sarstedt, "A New Criterion for Assessing Discriminant Validity in Variance-Based Structural Equation Modeling," *Journal of the Academy of Marketing Science* 43, no. 1 (2015): 115–135, https://doi.org/10.1007/s11747-014-0403-8.

[68] J. C. Nunnally, *Psychometric Theory*, 2nd ed. (McGraw-Hill, 1978).

[69] J. Henseler, C. M. Ringle, and M. Sarstedt, "Testing Measurement Invariance of Composites Using Partial Least Squares," *International Marketing Review* 33, no. 3 (2016): 405–431, https://doi.org/10.1108/imr-09-2014-0304.

[70] J. Soto, M. Orozco-Fontalvo, and S. A. Useche, "Public Transportation and Fear of Crime at BRT Systems: Approaching to the Case of Barranquilla (Colombia) through Integrated Choice and Latent Variable Models," *Transportation Research Part A: Policy and Practice* 155 (2022): 142–160, https://doi.org/10.1016/j.tra.2021.11.001.

[71] C. Sundling and V. Ceccato, "The Impact of Rail-Based Stations on Passengers' Safety Perceptions. A Systematic Review of International Evidence," *Transportation Research Part F: Traffic Psychology and Behaviour* 86 (2022): 99–120, https://doi.org/10.1016/j.trf.2022.02.011.

[72] A. Delbosc and G. Currie, "Modelling the Causes and Impacts of Personal Safety Perceptions on Public Transport Ridership," *Transport Policy* 24 (2012): 302–309, https://doi.org/10.1016/j.tranpol.2012.09.009.

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[73] L. A. B. Ouali, D. J. Graham, A. Barron, and M. Trompet, "Gender Differences in the Perception of Safety in Public Transport," *Journal of the Royal Statistical Society—Series A: Statistics in Society* 183, no. 3 (2020): 737–769, https://doi.org/10.1111/rssa.12558.

[74] J. Korn, "Riding Scared: Sexual Violence and Women's Mobility on Public Transportation in Santiago, Chile" (2018).

[75] G. Beirão and J. A. Sarsfield Cabral, "Understanding Attitudes Towards Public Transport and Private Car: A Qualitative Study," *Transport Policy* 14, no. 6 (2007): 478–489, https://doi.org/10.1016/j.tranpol.2007.04.009.

1409, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/atr/9606650, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License